Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., Pearce, A., Citro, C., Ameisen, E., Jones, A., Cunningham, H., Turner, N. L., McDougall, C., MacDiarmid, M., Freeman, C. D., Sumers, T. R., Rees, E., Batson, J., Jermyn, A., Carter, S., Olah, C., and Henighan, T. Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html.

Turnbull, O. M., Oglic, D., Croasdale-Wood, R., and Deane, C. M. p-IgGen: a paired antibody generative language model. *Bioinformatics*, 40(11):btae659, November 2024. ISSN 1367-4811. doi: 10.1093/bioinformatics/btae659. URL https://doi.org/10.1093/bioinformatics/btae659.

Wang, S. L., Quach, A., Parsan, N., and Yang, J. J. Enforcing Orderedness in SAEs to Improve Feature Consistency. September 2025. URL https://openreview.net/forum?id=0Y0BC0ldOQ.

Wang, Y., Lv, H., Teo, Q. W., Lei, R., Gopal, A. B., Ouyang, W. O., Yeung, Y.-H., Tan, T. J. C., Choi, D., Shen, I. R., Chen, X., Graham, C. S., and Wu, N. C. An explainable language model for antibody specificity prediction using curated influenza hemagglutinin antibodies. *Immunity*, 57(10):2453–2465.e7, October 2024. ISSN 1074-7613. doi: 10.1016/j.immuni.2024.07.022. URL https://www.cell.com/immunity/abstract/S1074-7613(24)00371-6. Publisher: Elsevier.

Zhang, L., Lu, Y., Wang, B., Li, F., and Zhang, Z. Sparse Auto-encoder with Smoothed $$l_1$$Regularization. *Neural Processing Letters*, 47(3):829–839, June 2018. ISSN 1573-773X. doi: 10.1007/s11063-017-9668-5. URL https://doi.org/10.1007/s11063-017-9668-5.

A. Supplementary Material

A.1. Case Study of Heavy J Gene Identity for IGHJ4

The functional importance of SAEs for studying LLMs lies in their ability to interrogate specific, domain-relevant concepts rather than an undefined set of all possible ones. Here, we examine gene identity, as it directly influences antibody binding affinity and specificity (Deng et al., 2025). Specifically, we pick IGHJ4 genes for our analysis, due to their widespread clinical significance underpinned by the fact that they are the most widely utilised J genes in our immune repertoire. That is, the majority of heavy chain antibodies within any given individual originate from the IGHJ4 germline gene.

Table 3. IGHJ4 feature statistics across p-IgGen layers. A latent is counted as a feature if it satisfies $F_1 > F_{1,\text{all}+} + 0.2$ for any tested activation threshold.

IGHJ	Layer	Features	Max F_1-score
IGHJ4	Layer 0	1	0.930
	Layer 1	4	0.949
	Layer 2	4	0.930
	Layer 3	1	0.752

For our analysis, we decided to focus on the final layer (Table 3).

First, we looked at the absolute positional activations of top IGHJ4 latents, based on F_1 score, across all the sequences in our validation set which had an IGHJ4 heavy J gene, and compared it to activations on specific IMGT positions. Whilst activations on absolute positions were distributed near the end of the heavy chain corresponding to the J region, the activations on IMGT positions were more consistent and concentrated. This implies that the model does not base its activation pattern on the absolute sequence length alone, but rather the underlying sequence alignment (Figure 4).

We then chose to investigate the specific residue identities on which the latents were activated. Based on the heavy J gene sequence alignments, the top two latents activated at IMGT positions 120 and 119, which are a Q and G, respectively. These are conserved across all human IGHJ genes. The third top latent activated on Y at position 117, which is unique for IGHJ4 (Scaviner et al., 1999). These results indicated that top latents encoded contextual information of the preceding residues.

Previous studies have highlighted how highly correlated features may be used to steer model outputs (Templeton et al., 2024; Simon & Zou, 2024). We attempted to steer on each identified latent to investigate how it affects model generation. We positively steered on each latent, which we hypothesised should increase the proportion of IGHJ4 in generated sequences. However, steering on these latents was unpredictable and did not consistently increase IGHJ4 proportions (Supplementary Figure 5).

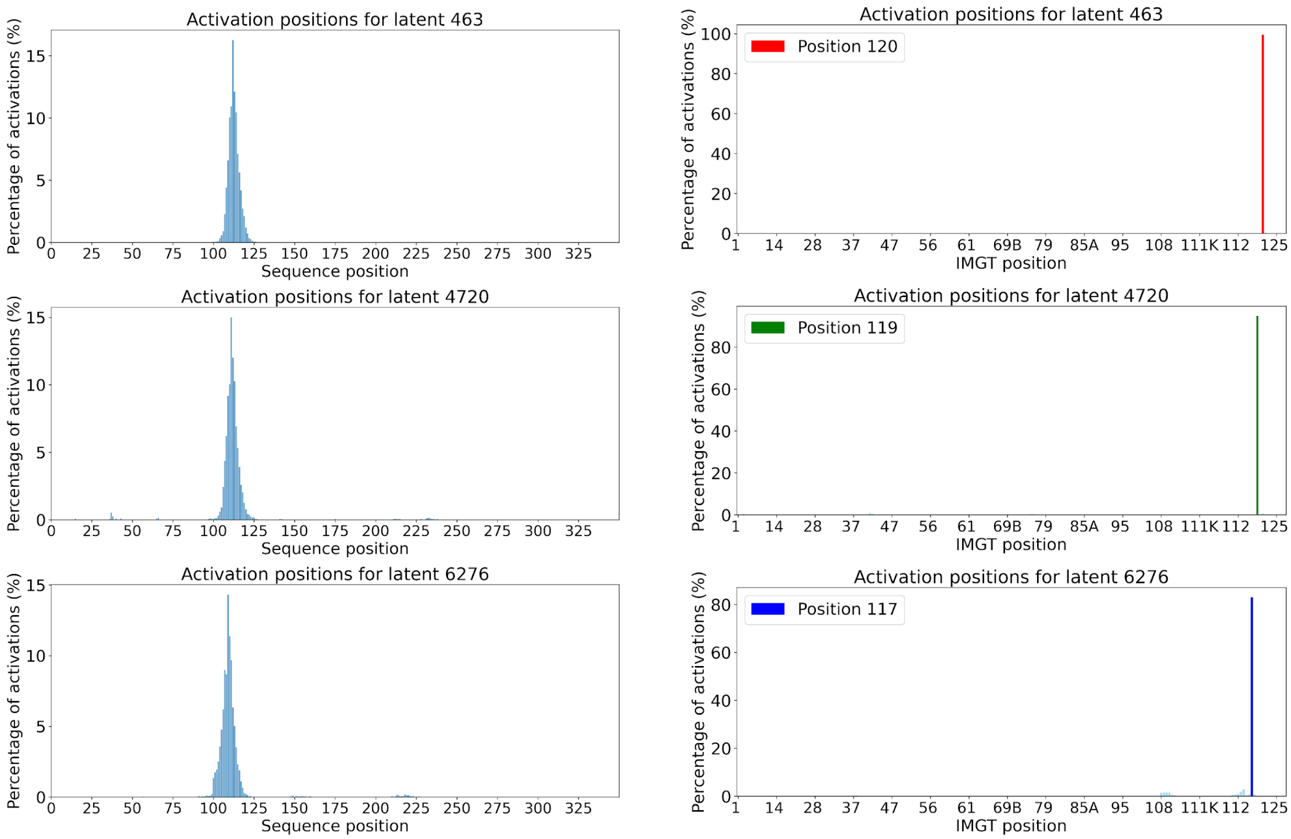
To check if this phenomenon was somehow exclusive for IGHJ4 and layer 3, we attempted to steer across all the layers for a number of different features for various gene identities, but were unable to predictably steer model generation [data not shown]. The lack of steerability may indicate how these features individually do not contribute to the gene identity, making them informative features when used for downstream predictions, but not for biasing model output.

This may be due to feature splitting (Chanin et al., 2024) which has been reported for TopK SAEs. Feature splitting refers to the phenomenon where higher-order features are broken down into specific contextual examples. In the case of text-based language models, 'math' may be split into 'algebra' and 'geometry'. These phenomena arise when enforcing sparsity in a dictionary consisting of hierarchical features. In this instance, if the identified latents correspond to only single residues within the J-domain, it essentially becomes a residue-level feature as opposed to a sequence-level feature. If the feature activates on a residue specific to the gene identity, it may be a good predictor for the gene identity, but not a steerable feature. This points to the possibility that several features together confer J gene identity, and that these features are likely correlated to each other. Hence, activating one but not the others does not necessarily result in a predictable shift in model performance.

The case study on IGHJ4 indicated that identified features retain biologically relevant context information. Most highly correlated features (based on LR correlation weight and F_1 score) tend to be residue-specific. Targeted approaches such as this cannot easily find abstract, higher-order features, assuming they are represented within the latent space to begin with. Concept-specific targeted feature identification might identify highly correlated features that are biologically informative. For instance, two of the three top features (463 and 4720) activate on conserved residues preceded by sequence motifs specific to the gene identity. The third, Latent 6276 activated on an IGHJ4-specific residue, which may explain why this feature can be used to accurately identify IGHJ4.

Highly predictive features may be correlated with other biologically informative features. To understand whether highly predictive features influence model behaviour, we tried to steer along these features to increase the proportion of IGHJ4 in generated sequences. This did not produce predictable results, making it difficult to interpret the contribution of each individual latent to model generation. Overall, TopK SAEs can identify features in targeted concept analysis which are intuitively interpretable, however, not necessarily steerable.

One of the reasons behind the lack of steerability of TopK latents may stem from feature splitting (Chanin et al., 2024), as highlighted earlier. As shown in this paper, a solution may be to use nested architectures which limit feature splitting, allowing greater steerability (Chanin et al., 2024).



(a) Absolute positional activations

(b) IMGT activations

Figure 4. Comparison of absolute positional (a) and IMGT (b) activations of top three IGHJ4 TopK latents ranked by F_1 score. Only one of the three pass teh 'feature-threshold' but the other latents still show an F_1 score greater than $F_{1,\text{all}+}$. The sequence/IMGT positions are shown on the x-axis. For the sequence positions, the amino acid sequences were end-padded to a constant length of 350. Percentage of total activations on any given position across validation IGHJ4 sequences is shown on the y-axis. The most frequent IMGT position for activation is highlighted for each latent. Latent activations show a distribution near the end of the heavy chain when aligned based on absolute sequence position. In contrast, latents demonstrate discrete activations when aligned based on IMGT numbering.

A.2. Supplementary Figures and Tables